

This Patient Group Direction (PGD) is intended only for registered healthcare professionals who have been named and authorised by their organisation to practice under it.

Patient Group Direction
for the administration of Doxycycline, Amoxicillin or Clarithromycin for the treatment of Acute Bacterial Bronchitis

Acute, Bacterial Bronchitis

From age 12 years onwards.

Assessment

Take a full history. In particular, ask about:

Onset of symptoms.

Duration of cough.

Frequency of cough.

Type of cough (dry or productive).

Precipitating or exacerbating factors.

Diurnal variation.

Associated symptoms – throat, chest, gastrointestinal.

Smoking history.

Environmental factors.

Occupational history.

Family history.

Recent respiratory tract infection.

Recent travel history.

Ask about other red flag clinical features, such as:

Haemoptysis.

Hoarseness.

Peripheral oedema with weight gain.

Prominent dyspnoea, especially at rest or at night.

Smokers aged over 45 years with a new cough, change in cough, or coexisting voice disturbance, and smokers aged 55–80 years who have a 30 pack-year smoking history and currently smoke or who have quit within the past 15 years.

Systemic symptoms, such as fever or weight loss.

Trouble swallowing.

Vomiting.

Use a validated tool to assess the severity of cough or quality of life (for example, the Leicester Cough questionnaire).

Consider comorbid conditions that may cause cough, such as:

Allergies — for more information, see the CKS topic on Allergic rhinitis.

Asthma — for more information, see the CKS topic on Asthma.

Chronic obstructive pulmonary disease (COPD) — for more information, see the CKS topic on Chronic obstructive pulmonary disease.

Bronchiectasis — for more information, see the CKS topic on Bronchiectasis.

Heart failure — for more information, see the CKS topic on Heart failure - chronic.

Gastroesophageal reflux disease — for more information, see the CKS topic on Dyspepsia - proven GORD.

Sleep apnoea syndrome — for more information, see the CKS topic on Obstructive sleep apnoea syndrome

Review the person's medication, such as use of angiotensin-converting enzyme (ACE) inhibitors, or sitagliptin.

Examine the person paying particular attention to the chest, heart, ears, pharynx, and nose.

Arrange investigations in primary care if appropriate, such as pulse oximetry (if the person is acutely unwell), peak expiratory flow rate (if asthma is known or suspected), pertussis serology (if whooping cough is suspected), C-reactive protein test (if pneumonia is suspected), and spirometry and chest X-ray (for example if there is chronic cough of uncertain cause).

Diagnosis

The differential diagnosis of cough is wide ranging and includes many diseases — the cause is determined by clinical features, medical history, elimination of alternative causes, and the response to targeted therapies where appropriate.

The duration of a cough at presentation can help guide diagnosis, however:

All coughs are acute at onset.

It is not possible to predict which cough will persist into sub-acute and chronic stages.

Recurrent acute coughs can be a manifestation of an undiagnosed chronic disease.

All chronic coughs start as sub-acute.

For people with acute cough (less than 3 weeks' duration):

Assess whether the person has clinical features of the most common cause:

Upper respiratory tract infection — suggested by cough with or without sputum, general malaise, and fever.

Pain and discharge may be localized to the nose, ears, throat, or sinuses. It is most commonly caused by a viral infection, such as a cold or flu.

Assess whether the person has clinical features of the following other possible causes, including:

Coronavirus disease 2019 (COVID-19) — a potentially severe acute respiratory infection. Signs and symptoms include cough, fever, and shortness of breath. For more information, see the CKS topic on Coronavirus - COVID 19.

Acute exacerbation of a pre-existing condition — for example, asthma, chronic obstructive pulmonary disease (COPD), bronchiectasis. For more information, see the CKS topics on Asthma, Chronic obstructive pulmonary disease and Bronchiectasis.

Acute bronchitis — suggested by cough with or without sputum, breathlessness, wheeze, or general malaise.

Crackles, if present, should clear with coughing. For more information, see the CKS topic on Chest infections - adult.

Pneumonia — suggested by at least one symptom of breathlessness, sputum, wheeze, or pleuritic pain, focal chest signs such as dull percussion note, bronchial breathing, or coarse crackles, plus at least one systemic feature such as fever or myalgia. There may be signs of an associated pleural effusion. For more information, see the CKS topic on Chest infections - adult.

Pneumothorax/tension pneumothorax — suggested by sudden-onset pleuritic pain, breathlessness, reduced chest wall movements, and breath sounds. Tension pneumothorax can result in tracheal deviation away from the pneumothorax, tachycardia, hypotension, and collapse.

Pulmonary embolism (PE) — suggested by acute-onset breathlessness, pleuritic pain, haemoptysis, crackles, and sinus tachycardia. For more information, see the CKS topic on Pulmonary embolism.

For people with sub-acute cough (3–8 weeks' duration):

Assess whether the person has clinical features of the most common cause:

Post-infectious cough — for example, after infections with *Mycoplasma pneumonia* or *Bordetella pertussis*.

This is suggested by persistent dry cough that started with an obvious respiratory tract infection, and the person is systemically well with a normal respiratory examination.

For people with chronic cough (over 8 weeks' duration):

Assess whether the person has clinical features of the most common cause:

Upper airway cough syndrome (post-nasal drip) — suggested by frequent throat clearing, an unpleasant sensation in the throat, nasal congestion, or nasal discharge and sneezing. Upper airway abnormalities are a key component and seasonal and occupational triggers are risk factors in some people. On examination, there may be visible mucus and a cobblestone appearance to the posterior oropharyngeal wall and local upper airway structures.

Assess whether the person has clinical features of the following common causes:

Angiotensin-converting enzyme (ACE) inhibitor-induced cough — suggested by bouts of coughing associated with the perception of airway irritation occurring within hours to months of the first dose.

Asthma — suggested by wheeze, breathlessness, worsening symptoms at night, in the morning, or with exercise and exposure to allergens. Peak expiratory flow rate is reduced during an episode. The prevalence of asthma in people with chronic cough ranges from 24–29%. For more information, see the CKS topic on Asthma.

Eosinophilic bronchitis — suggested by normal chest examination, normal spirometry, and normal response to airway provocation test (only available to a specialist).

Gastro-oesophageal reflux disease — suggested by cough that is worse during or after eating, with talking, and with bending. For more information, see the CKS topic on Dyspepsia - proven GORD.

Smoking-related cough — suggested by dry cough that is worse in the mornings.

Other possible causes of cough include:

Bronchiectasis — suggested by cough with daily sputum production, progressive breathlessness, haemoptysis, non-pleuritic chest pain, and coarse crackles in early inspiration in the lower lung fields. For more information, see the CKS topic on Bronchiectasis.

Bronchitis — suggested by cough with or without sputum, breathlessness, wheeze, or general malaise. Crackles, if present, should clear with coughing. For more information, see the CKS topic on Chest infections - adult.

Chronic obstructive pulmonary disease — suggested by persistent progressive breathlessness usually associated with wheezing or chest tightness, hyperinflated chest, possibly with signs of right-sided heart failure such as ankle oedema and increased jugular venous pressure. For more information, see the CKS topic on Chronic obstructive pulmonary disease.

Cough-variant asthma — suggested by normal chest examination, normal spirometry, and hyper-responsiveness to airway provocation test (only available to a specialist).

Environmental or occupational causes — cough can be an isolated symptom as a result of exposure to triggers (for example, perfume, dust, chemicals, pollutants) or manifestation of a more significant disease. Cough may be worse on weekdays and better at weekends for occupational causes.

Foreign body aspiration — suggested by sudden-onset cough, stridor (upper airway) or reduced chest wall movement on the affected side, bronchial breathing, and reduced or diminished breath sounds (lower airway).

Heart failure — suggested by significant breathlessness, orthopnoea, and/or paroxysmal nocturnal dyspnoea. There may be peripheral oedema and raised jugular venous pressure. For more information, see the CKS topic on Heart failure - chronic.

Interstitial lung disease (ILD) — suggested by breathlessness and cough (often dry). Although breathlessness is typical, people often present with non-specific signs and symptoms. There may be fine-end inspiratory crackles (indicative of fibrosis), finger clubbing, cyanosis, and signs of right-sided heart failure.

When the ILD is caused by sensitization to allergens, there may be a history of recurrent episodes of flu-like illness following exposure.

When the ILD is associated with connective tissue disease, there may be symptoms such as joint pain.

Diagnosis is confirmed by spirometry with a restrictive pattern, chest X-ray or high resolution computed tomography (HRCT).

Lung cancer — suggested by haemoptysis, persistent chest and/or shoulder pain, breathlessness, weight loss, hoarseness, finger clubbing, and cervical or supraclavicular lymphadenopathy. For more information, see the CKS topic on Lung and pleural cancers - recognition and referral.

Obstructive sleep apnoea syndrome — suggested by cough associated with features such as excessive daytime somnolence and obesity. For more information, see the CKS topic on Obstructive sleep apnoea syndrome.

Pertussis (whooping cough) — suggested by cough lasting more than 14 days with paroxysms of coughing. There may be vomiting after coughing, or an inspiratory whoop. For more information, see the CKS topic on Whooping cough.

Pulmonary tuberculosis — suggested by sputum, breathlessness, haemoptysis, weight loss, fever, night sweats, anorexia, general malaise, and finger clubbing. This should be considered in all patients complaining of cough in endemic areas, regardless of cough duration. For more information, see the CKS topic on Tuberculosis.

Sitagliptin-induced cough — suggested by cough accompanied with rhinorrhoea, dyspnoea, or fatigue.

Somatic cough syndrome — the diagnosis of somatic cough disorder can only be made after an extensive evaluation has been performed that includes ruling out tic disorders and uncommon causes and the person meets the DSM-5 criteria for a somatic symptom disorder.

How should I manage a person with acute bronchitis?

Refer or seek specialist advice on further investigation and management, if a person with acute cough has any symptoms or signs suggesting a more serious condition (for example sepsis, a pulmonary embolism, or lung cancer).

For more information, see the CKS topics on Sepsis, Pulmonary embolism, and Lung and pleural cancers - recognition and referral.

Advise the person on self-care strategies such as adequate fluid intake and the use of paracetamol or ibuprofen for symptomatic relief.

For more information, see the CKS topics on Analgesia - mild-to-moderate pain and NSAIDs - prescribing issues.

Advise that some people may wish to try the following self-care treatments:

Honey.

Pelargonium (a herbal medicine).

Over-the-counter cough medicines containing guaifenesin (an expectorant).

Over-the-counter cough medicines containing cough suppressants (except codeine) if the person does not have a persistent cough or excessive secretions.

If the person smokes, advise them to stop. For more information, see the CKS topic on Smoking cessation.

Advise the person to seek medical help if symptoms worsen rapidly or significantly, do not improve after 3–4 weeks, or they become systemically very unwell.

Do not routinely offer an antibiotic to treat an acute cough associated with acute bronchitis in people who are not systemically very unwell or at higher risk of complications. Inform the person that:

Acute bronchitis is usually a self-limiting illness and the cough usually lasts about 3–4 weeks.

Antibiotics do not make a large difference to the duration of symptoms, only shortening cough duration by about half a day on average.

Adverse effects, including diarrhoea and nausea are possible with antibiotic treatment.

Unnecessary antibiotic prescriptions may result in antibiotic resistance.

Offer an immediate antibiotic prescription if the person is systemically very unwell.

Consider an immediate antibiotic prescription or a backup antibiotic prescription for a person at higher risk of complications, for example:

A pre-existing comorbid condition such as heart, lung, kidney, liver, or neuromuscular disease; immunosuppression; or cystic fibrosis.

Older than 65 years of age with two or more of the following, or older than 80 years with one or more of the following:

Hospital admission in the previous year.

Type 1 or type 2 diabetes mellitus.

History of congestive heart failure.

Current use of oral corticosteroids.

If a back-up antibiotic prescription is appropriate:

Reassure the person that antibiotics are not currently needed.

Advise the person to use the delayed prescription if symptoms get rapidly or significantly worse.

Advise the person to seek medical advice if symptoms get rapidly or significantly worse despite taking the antibiotic, or they become systemically very unwell.

If a C-reactive protein (CRP) test has been carried out, use the results to guide antibiotic prescribing as follows:

CRP less than 20 mg/L — do not routinely offer antibiotics.

CRP 20–100 mg/L — consider a delayed antibiotic prescription.

CRP greater than 100 mg/L — offer antibiotic therapy.

If antibiotics are indicated, for adults 18 years of age and older:

First-line choice is oral doxycycline: 200 mg on the first day, then 100 mg once a day for 4 days (5-day course in total). Note doxycycline is not the first-choice antibiotic for pregnant women.

Alternative first choices are oral:

Amoxicillin 500 mg three times a day for 5 days.

Clarithromycin 250 mg to 500 mg twice a day for 5 days.

Erythromycin (preferred in pregnant women) 250 mg to 500 mg four times a day or 500 mg to 1000 mg twice a day for 5 days.

If antibiotics are indicated, for young people aged 12–17 years:

First-line choice is oral amoxicillin: 500 mg three times a day for 5 days.

Alternative first choices are oral:

Clarithromycin 250 mg to 500 mg twice a day for 5 days.

Erythromycin 250 mg to 500 mg four times a day or 500 mg to 1000 mg twice a day for 5 days.

Doxycycline 200 mg on first day, then 100 mg once a day for 4 days (5-day course in total). Note doxycycline should not be given to young women who are pregnant.

Do not offer an oral or inhaled bronchodilator (for example salbutamol) or an oral or inhaled corticosteroid to a person with an acute cough associated with acute bronchitis unless they have an underlying airway disease such as asthma.

Do not offer a mucolytic (for example acetylcysteine or carbocisteine) to treat an acute cough associated with acute bronchitis.

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Patient Group Direction

for the supply, for the treatment of acute, bacterial bronchitis,
of:

Doxycycline

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	● Pharmacist registered and practicing with the GPhC ● Pharmacy technician registered and practicing with the GPhC
Training and assessment	● Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. ● All users must previously have used PGD's to supply medication ● Must work in compliance with the SOPs of their own employer ● All users must have appropriate indemnity in place to cover this service
Further training and responsibilities	● All users must be up to date with CPD requirements and appraisal ● All users must be up to date with MHRA and safety alerts with regards to medical products ● All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005.

PGD Indication	Acute bacterial bronchitis
Inclusion criteria	Age 12 years and over with diagnosis of acute bacterial bronchitis
Exclusion criteria	Hypersensitivity to the active substance or other tetracyclines, or to any of the excipients Pregnancy and breastfeeding Red flags listed in accompanying guidance
Cautions	Use in patients with impaired hepatic function: Doxycycline should be administered with caution to patients with hepatic impairment or those receiving potentially hepatotoxic drugs. Abnormal hepatic function has been reported rarely and has been caused by both the oral and parenteral administration of tetracyclines, including doxycycline. Use in patients with renal impairment: Excretion of doxycycline by the kidney is about 40%/72 hours in individuals with normal renal function. This percentage excretion may fall to a range as low as 1-5%/72 hours in individuals with severe renal insufficiency (creatinine clearance below 10ml/min). Studies have shown no significant difference in the serum half-life of doxycycline in individuals with normal and severely impaired renal function. Haemodialysis does not alter the serum half-life of doxycycline. The anti-anabolic action of the tetracyclines may cause an increase in blood urea. Studies to date indicate that this anti-anabolic effect does not occur with the use of Doxycycline in patients with impaired renal function. Serious skin reactions: Serious skin reactions, such as exfoliative dermatitis, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis, and drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported in patients receiving doxycycline. If serious skin reactions occur, doxycycline should be discontinued immediately, and appropriate therapy should be instituted. Photosensitivity Photosensitivity manifested by an exaggerated sunburn reaction has been observed in some individuals taking tetracyclines, including doxycycline. Patients likely to be exposed to direct sunlight or ultraviolet light should be advised that this reaction can occur with tetracycline drugs and treatment should be discontinued at the first evidence of skin erythema. Photoonycholysis has also been reported in patients receiving doxycycline. Benign intracranial hypertension Bulging fontanelles in infants have been reported in individuals receiving tetracyclines. Benign intracranial hypertension (pseudotumor cerebri) has been associated with the use of tetracyclines including doxycycline. Benign intracranial hypertension (pseudotumor cerebri) is usually transient, however cases of permanent visual loss secondary to benign intracranial hypertension (pseudotumor cerebri) have been reported with tetracyclines including doxycycline. If visual disturbance occurs during treatment, prompt ophthalmologic evaluation is warranted. Since intracranial pressure can remain elevated for weeks after drug cessation patients should be monitored until they stabilize. Concomitant use of isotretinoin or other systemic retinoids and doxycycline should be avoided because isotretinoin is also known to cause benign intracranial hypertension (pseudotumor cerebri). Microbiological overgrowth: The use of antibiotics may occasionally result in the overgrowth of non-susceptible organisms including Candida. If a resistant organism appears, the antibiotic should be discontinued and appropriate therapy instituted. Pseudomembranous colitis has been reported with nearly all antibacterial agents, including doxycycline, and has ranged in severity from mild to life-threatening. It is important to consider this diagnosis in patients who present with diarrhoea

	subsequent to the administration of antibacterial agents. Clostridium difficile associated diarrhoea (CDAD) has been reported with use of nearly all antibacterial agents, including doxycycline, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of C. Difficile. C. difficile produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of C. Difficile cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhoea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents. Oesophagitis: Instances of oesophagitis and oesophageal ulcerations have been reported in patients receiving capsule and tablet forms of drugs in the tetracycline class, including doxycycline. Most of these patients took medications immediately before going to bed or with inadequate amounts of fluid. Porphyria: There have been rare reports of porphyria in patients receiving tetracyclines. Myasthenia gravis: Due to a potential for weak neuromuscular blockade, care should be taken in administering tetracyclines to patients with myasthenia gravis. Systemic lupus erythematosus: Tetracyclines can cause exacerbation of SLE. Methoxyflurane: Caution is advised in administering tetracyclines with methoxyflurane.
Actions if patient is excluded or declines treatment	Advise on alternative treatment options and how these can be accessed. Document any advice given and the decision reached. Inform or refer to the GP as appropriate

Details of the medicine

Name, form and strength of medicine	Each hard capsule contains doxycycline hyclate corresponding to 100 mg doxycycline.
Legal category	POM
Storage	Do not store above 25°C. Keep in the original package. Store in a dry place.
Route/method of administration	The capsules should be swallowed with plenty of water in either the sitting or standing position and well before retiring at night to reduce the risk of oesophageal irritation and ulceration. If gastric irritation occurs, it is recommended that Doxycycline capsules be given with food or milk. Studies indicate that the absorption of doxycycline is not notably influenced by simultaneous ingestion of food or milk.
Dose and frequency	Adults and children aged 12 years to less than 18 years: The usual dosage of Doxycycline capsules for the treatment of acute infections in adults and children aged 12 years to less than 18 years is 200 mg on the first day (as a single dose) followed by 100 mg daily for a further four days (total of five days treatment).
Quantity to be administered and/or supplied	6 capsules
Maximum or minimum	5 days

treatment period	
Adverse effects	Commonly reported adverse effects are headache, nausea and vomiting and photosensitivity reactions. Refer to SmPC.

Yellow card reporting	Healthcare professionals and individuals/carers should report suspected adverse reactions to the Medicines and Healthcare Products Regulatory Agency (MHRA) using the Yellow Card reporting scheme on https://yellowcard.mhra.gov.uk/ Any adverse reaction to a medication should be documented in Accordance with locally agreed procedures in the individual's record and the individual's GP should be informed.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP ● name of healthcare practitioner ● name and brand of medication ● date of supply dose, form and route quantity supplied ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken supplied via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. A record of all individuals receiving treatment under this PGD should also be kept for audit purposes as follows: ● Adults aged 18 years and over - keep records for 8 years ● Children aged under 18 years -keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medication.
Follow-up advice to be given to patient or carer	To seek medical advice if symptoms worsen rapidly or significantly, do not improve in 3–4 weeks, or they become systemically very unwell.

Key references

● NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions. Published March 2017 https://www.nice.org.uk/guidance/mg2 ● NICE MPG2 Patient group directions: competency framework for health professionals using patient group directions. Updated March 2017 https://www.nice.org.uk/guidance/mpg2/resources ● NICE CKS Guidance ● British National Formulary (BNF) Key information on the selection, prescribing, dispensing and administration of medicines BNF (British National Formulary) NICE ● Summary of Product Characteristics

Agreement to practise by the Registered Healthcare Professional

By signing this PGD you are confirming that you agree to its contents and that you will work within it. PGDs do not remove inherent professional obligations or accountability. It is the responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

Name and address of pharmacy premises / clinic premises to which this PGD relates			
Name of healthcare professional	Job title	Registration number	Signature
Valid from date		Expiry date	
12/08/2025		12/08/2027	

PGD adoption and authorisation by the employer/clinical lead

The employer has ensured that the person using this PGD, has the knowledge and skills to safely provide the service which will encompass the supply or administration of a medication or vaccine.

This section must be signed by the superintendent or clinical lead responsible for this service.

Name of superintendent / clinical lead	Job title & name of organisation	Signature	Date

Name of professional group signatory	Job title & name of organisation	Signature	Date

Signed on behalf of Get Real Health

Name	Qualification	Signature	Date
Nitin Shori	Medical Doctor GMC: 6047293		12/8/2025
Chris Pilkington	Pharmacist GPhC: 2046322		12/8/2025

Change history

Version number	Change details	Date
001	Development & issue of new PGD	25 th June 2025

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Patient Group Direction

for the supply, for the treatment of acute bacterial bronchitis,
of:

Amoxicillin

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	● Pharmacist registered and practicing with the GPhC ● Pharmacy technician registered and practicing with the GPhC
Training and assessment	● Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. ● All users must be familiar with the SPC for the products as well as BNF advice and any applicable national guidelines ● All users must previously have used PGD's to supply medication ● Must work in compliance with the SOPs of their own employer ● All users must have appropriate indemnity in place to cover this service
Further training and responsibilities	● All users must be up to date with CPD requirements and appraisal ● All users must be up to date with MHRA and safety alerts with regards to medical products ● All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005.

PGD Indication	Acute, bacterial bronchitis
Inclusion criteria	Age 12 years and over with diagnosis of acute bacterial bronchitis
Exclusion criteria	Hypersensitivity to the active substance, to any of the penicillins or to any of the excipients. History of a severe immediate hypersensitivity reaction (e.g. anaphylaxis) to another betalactam agent (e.g. a cephalosporin, carbapenem or monobactam). Pregnancy Red flags listed in accompanying guidance
Cautions	Hypersensitivity reactions: Before initiating therapy with amoxicillin, careful enquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins or other beta-lactam agents. Serious and occasionally fatal hypersensitivity (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients on penicillin therapy. Hypersensitivity reactions can also progress to Kounis syndrome, a serious allergic reaction that can result in myocardial infarction. These reactions are more likely to occur in individuals with a history of penicillin hypersensitivity and in atopic individuals. If an allergic reaction occurs, amoxicillin therapy must be discontinued and appropriate alternative therapy instituted. Convulsions Convulsions may occur in patients with impaired renal function or in those receiving high doses or in patients with predisposing factors (e.g. history of seizures, treated epilepsy or meningeal disorders). Renal impairment In patients with renal impairment, the dose should be adjusted according to the degree of impairment. Skin reactions: The occurrence at the treatment initiation of a feverish generalised erythema associated with pustula may be a symptom of acute generalised exanthemous pustulosis. This reaction requires amoxicillin discontinuation and contra-indicates any subsequent administration. Amoxicillin should be avoided if infectious mononucleosis is suspected since the occurrence of a morbilliform rash has been associated with this condition following the use of amoxicillin. Overgrowth of non-susceptible microorganisms: Prolonged use may occasionally result in overgrowth of non-susceptible organisms. Antibiotic-associated colitis has been reported with nearly all antibacterial agents and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea during, or subsequent to, the administration of any antibiotics. Should antibiotic-associated colitis occur, amoxicillin should immediately be discontinued, a physician consulted and an appropriate therapy initiated. Anti-peristaltic medicinal products are contra-indicated in this situation.

	Anticoagulants Prolongation of prothrombin time has been reported rarely in patients receiving amoxicillin. Appropriate monitoring should be undertaken when anticoagulants are prescribed concomitantly. Adjustments in the dose of oral anticoagulants may be necessary to maintain the desired level of anticoagulation Drug-induced enterocolitis syndrome (DIES) Drug-induced enterocolitis syndrome (DIES) has been reported mainly in children receiving amoxicillin. DIES is an allergic reaction with the leading symptom of protracted vomiting (1-4 hours after drug intake) in the absence of allergic skin or respiratory symptoms. Further symptoms could comprise abdominal pain, diarrhoea, hypotension or leucocytosis with neutrophilia. There have been severe cases including progression to shock. Crystalluria In patients with reduced urine output, crystalluria (including acute renal injury) has been observed very rarely, predominantly with parenteral therapy. During the administration of high doses of amoxicillin, it is advisable to maintain adequate fluid intake and urinary output in order to reduce the possibility of amoxicillin crystalluria. In patients with bladder catheters, a regular check of patency should be maintained. Interference with diagnostic tests Elevated serum and urinary levels of amoxicillin are likely to affect certain laboratory tests. Due to the high urinary concentrations of amoxicillin, false positive readings are common with chemical methods. It is recommended that when testing for the presence of glucose in urine during amoxicillin treatment, enzymatic glucose oxidase methods should be used. The presence of amoxicillin may distort assay results for oestriol in pregnant women. Pregnancy: Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. Limited data on the use of amoxicillin during pregnancy in humans do not indicate an increased risk of congenital malformations. Amoxicillin may be used in pregnancy when the potential benefits outweigh the potential risks associated with treatment. Breastfeeding: Amoxicillin is excreted into breast milk in small quantities with the possible risk of sensitisation. Consequently, diarrhoea and fungus infection of the mucous membranes are possible in the breast-fed infant, so that breast-feeding might have to be discontinued. Amoxicillin should only be used during breast-feeding after benefit/risk assessment by the physician in charge. No studies on the effects on the ability to drive and use machines have been performed. However, undesirable effects may occur (e.g. allergic reactions, dizziness, convulsions), which may influence the ability to drive and use machines
Actions if patient is excluded or declines	Advise on alternative treatment options and how these can be accessed. Document any advice given and the decision reached. Inform or refer to the GP

treatment	as appropriate
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Details of the medicine

Name, form and strength of medicine	Each hard capsule contains amoxicillin trihydrate equivalent to 500 mg amoxicillin.		
Legal category	POM		
Storage	Do not store above 25°C. Store in the original package and keep containers tightly closed.		
Route/method of administration	Amoxicillin capsules are for oral use. Absorption of Amoxicillin capsules is unimpaired by food. Swallow with water without opening capsule.		
Dose and frequency	Child 12–17 years 500 mg 3 times a day for 5 days.		
	Adult 500 mg 3 times a day for 5 days.		
	Renal impairment		
	GFR (ml/min)	Adults and children ≥ 40 kg	Children < 40 kg [#]
	greater than 30	no adjustment necessary	no adjustment necessary
	10 to 30	maximum 500 mg twice daily	15 mg/kg given twice daily (maximum 500 mg twice daily)
	less than 10	maximum 500 mg/day.	15 mg/kg given as a single dose (maximum 500 mg daily)
Quantity to be administered and/or supplied	15 capsules (or as appropriate for adjusted dose)		
Maximum or minimum treatment period	5 days		
Adverse effects	The most commonly reported adverse drug reactions (ADRs) are diarrhoea, nausea and skin rash. Refer to SmPC.		

Yellow card reporting	Healthcare professionals and individuals/carers should report suspected adverse reactions to the Medicines and Healthcare Products Regulatory Agency (MHRA) using the Yellow Card reporting scheme on https://yellowcard.mhra.gov.uk/ Any adverse reaction to a medication should be documented in accordance with locally agreed procedures in the individual's record and the individual's GP should be informed.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP ● name of healthcare practitioner ● name and brand of medication

	● date of supply dose, form and route quantity supplied ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken supplied via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. A record of all individuals receiving treatment under this PGD should also be kept for audit purposes as follows: ● Adults aged 18 years and over - keep records for 8 years ● Children aged under 18 years -keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)
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Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medication.
Follow-up advice to be given to patient or carer	To seek medical advice if symptoms worsen rapidly or significantly, do not improve in 3–4 weeks, or they become systemically very unwell.

Key references

● NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions. Published March 2017 https://www.nice.org.uk/guidance/mpg2 ● NICE MPG2 Patient group directions: competency framework for health professionals using patient group directions. Updated March 2017 https://www.nice.org.uk/guidance/mpg2/resources ● NICE CKS Guidance ● British National Formulary (BNF) Key information on the selection, prescribing, dispensing and administration of medicines BNF (British National Formulary) NICE ● Summary of Product Characteristics

Agreement to practise by the Registered Healthcare Professional

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Name and address of pharmacy premises / clinic premises to which this PGD relates

Name of healthcare professional	Job title	Registration number	Signature
Valid from date		Expiry date	
12/08/2025		12/08/2027	

PGD adoption and authorisation by the employer/clinical lead

The employer has ensured that the person using this PGD, has the knowledge and skills to safely provide the service which will encompass the supply or administration of a medication or vaccine.

This section must be signed by the superintendent or clinical lead responsible for this service.

Name of superintendent / clinical lead	Job title & name of organisation	Signature	Date

Name of professional group signatory

Job title & name of organisation

Signature

Date

Signed on behalf of Get Real Health

Name	Qualification	Signature	Date
Nitin Shori	Medical Doctor GMC: 6047293		12/8/2025
Chris Pilkington	Pharmacist GPhC: 2046322		12/8/2025

Change history

Version number	Change details	Date
001	Development & issue of new PGD	25 th June 2025

This Patient Group Direction (PGD) is intended only for registered healthcare professionals who have been named and authorised by their organisation to practice under it.

Healthcare professionals must work within their own competence. This PGD does not remove professional obligations or accountability.

Patient Group Direction

for the supply, for the treatment of acute, bacterial bronchitis,
of:

Clarithromycin

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	● Pharmacist registered and practicing with the GPhC ● Pharmacy technician registered and practicing with the GPhC
Training and assessment	● Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. ● All users must be familiar with the SPC for the products as well as BNF advice and any applicable national guidelines ● All users must previously have used PGD's to supply medication ● Must work in compliance with the SOPs of their own employer ● All users must have appropriate indemnity in place to cover this service
Further training and responsibilities	● All users must be up to date with CPD requirements and appraisal ● All users must be up to date with MHRA and safety alerts with regards to medical products ● All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005.

PGD Indication	Acute, bacterial bronchitis
Inclusion criteria	Age 12 years and over with diagnosis of acute bacterial bronchitis
Exclusion criteria	Pneumonia. Pregnancy and breastfeeding Patients taking oral anticoagulants such as warfarin or direct oral anticoagulants (DOACs). Clarithromycin is contraindicated in patients with known hypersensitivity to the active substance clarithromycin, to other macrolide antibiotic drugs or to any of the excipients Concomitant administration of clarithromycin and ergot alkaloids (e.g., ergotamine or dihydroergotamine) is contraindicated, as this may result in ergot toxicity Concomitant administration of clarithromycin and oral midazolam is contraindicated Concomitant administration of clarithromycin and lomitapide is contraindicated Concomitant administration of clarithromycin and any of the following drugs is contraindicated: astemizole, cisapride, domperidone, pimozone and terfenadine as this may result in QT prolongation and cardiac arrhythmias, including ventricular tachycardia, ventricular fibrillation and torsades de pointes Clarithromycin should not be given to patients with history of QT prolongation (congenital or documented acquired QT prolongation) or ventricular cardiac arrhythmia, including torsades de pointes Concomitant administration with ticagrelor, ivabradine or ranolazine is contraindicated. Clarithromycin should not be used concomitantly with HMG-CoA reductase inhibitors (statins) that are extensively metabolised by CYP3A4 (lovastatin or simvastatin), due to the increased risk of myopathy, including rhabdomyolysis As with other strong CYP3A4 inhibitors, Clarithromycin should not be used in patients taking colchicine Clarithromycin should not be given to patients with electrolyte disturbances (hypokalaemia or hypomagnesaemia, due to the risk of prolongation of the QT interval). Clarithromycin should not be used in patients who suffer from severe hepatic failure in combination with renal impairment. Red flags listed in accompanying guidance
Cautions	There are no data on the effect of clarithromycin on the ability to drive or use machines. The potential for dizziness, vertigo, confusion and disorientation, which may occur with the medication, should be taken into account before patients drive or use machines. Clarithromycin is principally metabolised by the liver. Therefore, caution should be exercised in administering this antibiotic to patients with impaired hepatic function. Caution should also be exercised when administering clarithromycin to patients with moderate to severe renal impairment

	Hepatic dysfunction, including increased liver enzymes, and hepatocellular and/or cholestatic hepatitis, with or without jaundice, has been reported with clarithromycin. This hepatic dysfunction may be severe and is usually reversible. Cases of fatal hepatic failure have been reported. Some patients may have had pre-existing hepatic disease or may have been taking other hepatotoxic medicinal products. Patients should be advised to stop treatment and contact their doctor if signs and symptoms of hepatic disease develop, such as anorexia, jaundice, dark urine, pruritus, or tender abdomen. Pseudomembranous colitis has been reported with nearly all antibacterial agents, including macrolides, and may range in severity from mild to life-threatening. Clostridioides difficile-associated diarrhoea (CDAD) has been reported with use of nearly all antibacterial agents including clarithromycin, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon, which may lead to overgrowth of C. Difficile. CDAD must be considered in all patients who present with diarrhoea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents. Therefore, discontinuation of clarithromycin therapy should be considered regardless of the indication. Microbial testing should be performed and adequate treatment initiated. Drugs inhibiting peristalsis should be avoided. There have been post-marketing reports of colchicine toxicity with concomitant use of clarithromycin and colchicine, especially in the elderly, some of which occurred in patients with renal insufficiency. Deaths have been reported in some such patients. Concomitant administration of clarithromycin and colchicine is contraindicated. Caution is advised regarding concomitant administration of clarithromycin and triazolobenzodiazepines, such as triazolam, and intravenous or oromucosal midazolam. Cardiovascular Events Prolongation of the QT interval, reflecting effects on cardiac repolarisation imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in patients treated with macrolides including clarithromycin. Due to increased risk of QT prolongation and ventricular arrhythmias (including torsades de pointes), the use of clarithromycin is contraindicated in patients taking any of astemizole, cisapride, domperidone, pimozide and terfenadine; in patients who have electrolyte disturbances such as hypomagnesaemia or hypokalaemia; and in patients with a history of QT prolongation or ventricular cardiac arrhythmia. Carefully consider the balance of benefits and risks before prescribing clarithromycin for any patients taking hydroxychloroquine or chloroquine, because of the potential for an increased risk of cardiovascular events and cardiovascular mortality. Furthermore, clarithromycin should be used with caution in the following: • Patients with coronary artery disease, severe cardiac insufficiency, conduction disturbances or clinically relevant bradycardia; • Patients concomitantly taking other medicinal products associated with QT prolongation other than those which are contraindicated. Epidemiological studies investigating the risk of adverse cardiovascular outcomes with macrolides have shown variable results. Some observational studies have identified a rare short-term risk of arrhythmia, myocardial infarction and cardiovascular mortality associated with macrolides including clarithromycin. Consideration of these findings should be balanced with treatment benefits when prescribing clarithromycin. In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCAR) (e.g. Acute generalised exanthematous pustulosis (AGEP), Stevens-Johnson Syndrome, toxic epidermal
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	necrolysis and Drug Rash with Eosinophilia and Systemic Symptoms (DRESS), clarithromycin therapy should be discontinued immediately and appropriate treatment should be urgently initiated. Clarithromycin should be used with caution when administered concurrently with medications that induce the cytochrome CYP3A4 enzyme HMG-CoA Reductase Inhibitors (statins): Concomitant use of clarithromycin with lovastatin or simvastatin is contraindicated. Caution should be exercised when prescribing clarithromycin with other statins. Rhabdomyolysis has been reported in patients taking clarithromycin and statins. Patients should be monitored for signs and symptoms of myopathy. Oral hypoglycaemic agents/Insulin: The concomitant use of clarithromycin and oral hypoglycaemic agents (such as sulfonylureas) and/or insulin can result in significant hypoglycaemia. Careful monitoring of glucose is recommended. Attention should also be paid to the possibility of cross resistance between clarithromycin and other macrolide drugs, as well as lincomycin and clindamycin
Actions if patient is excluded or declines treatment	Advise on alternative treatment options and how these can be accessed. Document any advice given and the decision reached. Inform or refer to the GP as appropriate

Details of the medicine

Name, form and strength of medicine	Each film-coated tablet contains 250 mg or 500mg of clarithromycin
Legal category	POM
Storage	Do not store above 25°C. Store in the original package and keep containers tightly closed.
Route/method of administration	Clarithromycin tablets should be swallowed whole without crushing or chewing. Clarithromycin may be given without regard to meals as food does not affect the extent of bioavailability.
Dose and frequency	Adults: The usual dose is 250 mg twice daily although this may be increased to 500mg twice daily in severe infections. The usual duration of treatment is 6 to 14 days. Children older than 12 years: As for adults. In patients with renal impairment with creatinine clearance less than 30 mL/min, the dosage of clarithromycin should be reduced by one-half, i.e. 250 mg once daily, or 250 mg twice daily in more severe infections.
Quantity to be administered and/or supplied	10 tablets
Maximum or minimum treatment period	5 days
Adverse effects	The most frequent and common adverse reactions related to clarithromycin therapy for both adult and paediatric populations are abdominal pain, diarrhoea, nausea, vomiting and taste perversion. Refer to SmPC.

Yellow card reporting	Healthcare professionals and individuals/carers should report suspected adverse reactions to the Medicines and Healthcare Products Regulatory Agency (MHRA) using the Yellow Card reporting scheme on https://yellowcard.mhra.gov.uk/ Any adverse reaction to a medication should be documented in accordance with locally agreed procedures in the individual's record and the individual's GP should be informed.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP ● name of healthcare practitioner ● name and brand of medication ● date of supply dose, form and route quantity supplied ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken supplied via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. A record of all individuals receiving treatment under this PGD should also be kept for audit purposes as follows: ● Adults aged 18 years and over - keep records for 8 years ● Children aged under 18 years -keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

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- NICE MPG2 Patient group directions: competency framework for health professionals using patient group directions. Updated March 2017 <https://www.nice.org.uk/guidance/ MPG2/resources>
- NICE CKS Guidance
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Name of professional group signatory

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Name	Qualification	Signature	Date
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Change history

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Appendix 1: Decision Aid

History	
Onset of symptoms.	
Duration of cough.	
Frequency of cough.	
Type of cough (dry or productive).	
Precipitating or exacerbating factors.	
Diurnal variation.	
Associated symptoms – throat, chest, gastrointestinal.	
Smoking history.	
Environmental factors.	
Occupational history.	
Family history.	
Recent respiratory tract infection.	
Recent travel history.	

Observations	
Temperature	
Blood pressure	
Pulse	
Respiratory rate	
SpO2	
Level of consciousness	

History	Yes	No
Current COPD/asthma*		
Smokers aged over 45 years with a new cough, change in cough, or coexisting voice disturbance		
Smokers aged 55–80 years who have a 30 pack-year smoking history and currently smoke or who have quit within the past 15 years		
Cough lasting more than 3 weeks		
Pregnancy/breastfeeding		
Immunocompromised		

Presentation	Yes	No
Haemoptysis		
Vomiting		
Peripheral oedema with weight gain		
Persistent hoarseness		
Severely systemically unwell		
Prominent dyspnoea, especially at rest or at night		
Dysphagia		
pO ₂ below 97		

Examination	Yes	No
Absence of normal breath sounds		
Stridor		
Unilateral variation in lung sounds		
Cyanosis of lips/extremities		
Auxiliary muscle use		

If the answer to any of the questions is yes, refer to an appropriate healthcare professional